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Java SE 27 & JDK 27

Class BufferedOutputStream

java.lang.Object
  java.io.OutputStream
    java.io.FilterOutputStream
      java.io.BufferedOutputStream

All Implemented Interfaces:

Closeable, Flushable, AutoCloseable

```
public class BufferedOutputStream
extends FilterOutputStream
```

The class implements a buffered output stream. By setting up such an output stream, an application can write bytes to the underlying output stream without necessarily causing a call to the underlying system for each byte written.

API Note:

Once wrapped in a `BufferedOutputStream`, the underlying `OutputStream` should not be used directly nor wrapped with another stream.

Since:

1.0

Field Summary

Fields

Modifier and Type	Field	Description
`protected byte[]`	`buf`	The internal buffer where data is stored.
`protected int`	`count`	The number of valid bytes in the buffer.

Fields declared in class FilterOutputStream

`out`

Constructor Summary

Constructors

Constructor	Description
`BufferedOutputStream(OutputStream out)`	

	Creates a new buffered output stream to write data to the specified underlying output stream.
BufferedOutputStream(OutputStream out, int size)	Creates a new buffered output stream to write data to the specified underlying output stream with the specified buffer size.

Method Summary

All Methods | Instance Methods

Modifier and Type	Method	Description
void	**flush**()	Flushes this buffered output stream.
void	**write**(byte[] b, int off, int len)	Writes `len` bytes from the specified byte array starting at offset `off` to this buffered output stream.
void	**write**(int b)	Writes the specified byte to this buffered output stream.

Methods declared in class FilterOutputStream

close, write

Methods declared in class OutputStream

nullOutputStream

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Field Details

buf

```
protected byte[] buf
```

The internal buffer where data is stored.

count

```
protected int count
```

The number of valid bytes in the buffer. This value is always in the range `0` through `buf.length`; elements `buf[0]` through `buf[count-1]` contain valid byte data.

Constructor Details

BufferedOutputStream

```
public BufferedOutputStream(OutputStream out)
```

Creates a new buffered output stream to write data to the specified underlying output stream.

Parameters:

`out` - the underlying output stream.

BufferedOutputStream

```
public BufferedOutputStream(OutputStream out,
                            int size)
```

Creates a new buffered output stream to write data to the specified underlying output stream with the specified buffer size.

Parameters:

`out` - the underlying output stream.

`size` - the buffer size.

Throws:

`IllegalArgumentException` - if size <= 0.

Method Details

write

```
public void write(int b)
           throws IOException
```

Writes the specified byte to this buffered output stream.

Overrides:

`write` in class `FilterOutputStream`

Parameters:

`b` - the byte to be written.

Throws:

`IOException` - if an I/O error occurs.

write

```
public void write(byte[] b,
                  int off,
                  int len)
           throws IOException
```

Writes `len` bytes from the specified byte array starting at offset `off` to this buffered output stream.

Ordinarily this method stores bytes from the given array into this stream's buffer, flushing the buffer to the underlying output stream as needed. If the requested length is at least as large as this stream's buffer, however, then this method will flush the buffer and write the bytes directly to the underlying output stream. Thus redundant `BufferedOutputStream`s will not copy data unnecessarily.

Overrides:

`write` in class `FilterOutputStream`

Parameters:

`b` - the data.

`off` - the start offset in the data.

`len` - the number of bytes to write.

Throws:

`IOException` - if an I/O error occurs.

`IndexOutOfBoundsException` - If `off` is negative, `len` is negative, or `len` is greater than `b.length - off`

See Also:

`FilterOutputStream.write(int)`

flush

```
public void flush()
           throws IOException
```

Flushes this buffered output stream. This forces any buffered output bytes to be written out to the underlying output stream.

Specified by:

`flush` in interface `Flushable`

Overrides:

`flush` in class `FilterOutputStream`

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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